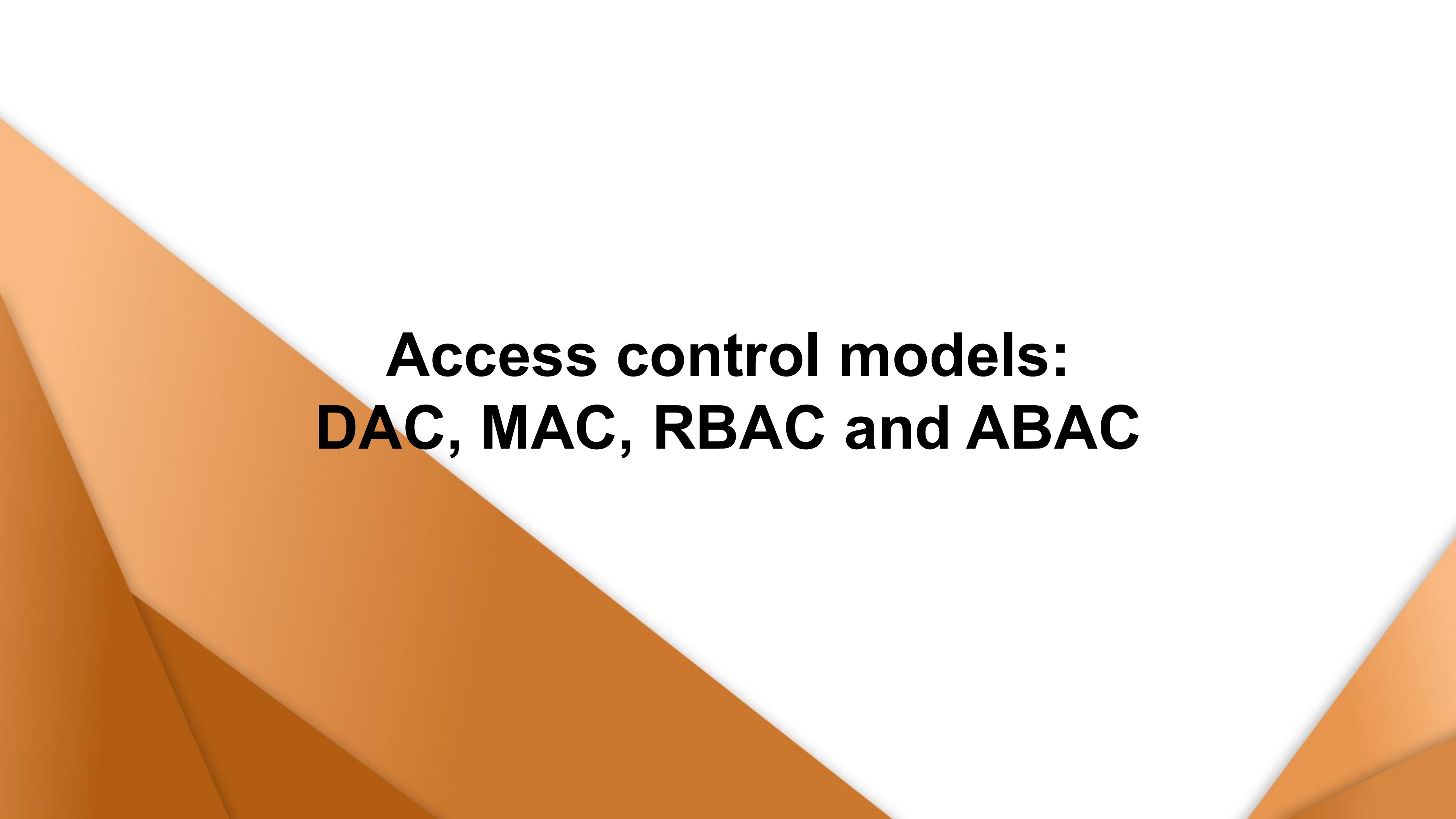


The background features abstract, overlapping geometric shapes in various shades of orange and brown, creating a modern, layered effect. The shapes are primarily triangular and polygonal, with some having soft shadows to give a sense of depth.

Access control models: DAC, MAC, RBAC and ABAC

Learning objectives

- **Define access control**
- **Explain access control is critical for cybersecurity**
- **Describe purpose of each model: DAC, MAC, RBAC, ABAC and how they compare**

What is access control?

Access control = rules for who can access what

- **Prevents unauthorised access**
- **Protects systems, files and data**
- **Enforces organisational security policies**

Discretionary Access Control (DAC)

- **Access is controlled by the resource owner**
- **Users can grant access to others**
- **Flexible but less secure**
- **Common in personal computers and file sharing**

Mandatory Access Control (MAC)

- **Access is based on system-wide security labels**
- **Users cannot change permissions**
- **Common in military and government systems**
- **Very secure but inflexible**

Role-Based Access Control (RBAC)

- **Access is based on user's role in the organisation**
- **Roles group permissions (e.g. Manager, HR, IT)**
- **Easier to manage at scale**
- **Common in businesses and enterprise systems**

Attribute-Based Access Control (ABAC)

- **Most flexible model**
- **Access based on attributes (user, resource, environment)**
- **Supports complex rules (e.g. location, time)**
- **Powerful, more complex to set up**
- **Used in cloud and large-scale systems**

Comparing access control models

Model	Who Controls Access	Best For	Flexibility	Security
DAC	User	Small teams	High	Low
MAC	System	Government or military	Low	High
RBAC	Based on Role	Enterprises	Medium	Medium
ABAC	Based on Attributes	Cloud or large scale	High	High

Why access control matters in cybersecurity

- **Protects sensitive data**
- **Minimises insider threats**
- **Supports compliance (e.g. GDPR, HIPAA)**
- **Reduces risk of accidental or malicious access**

A quick recap

- **Access control = managing who can access what**
- **Four main models: DAC, MAC, RBAC, ABAC**
- **Each has pros, cons and ideal uses**

Next: an introduction to application security

You will learn:

- **How to build and protect secure software systems**
- **Common threats to applications**
- **Secure coding principles and best practices**